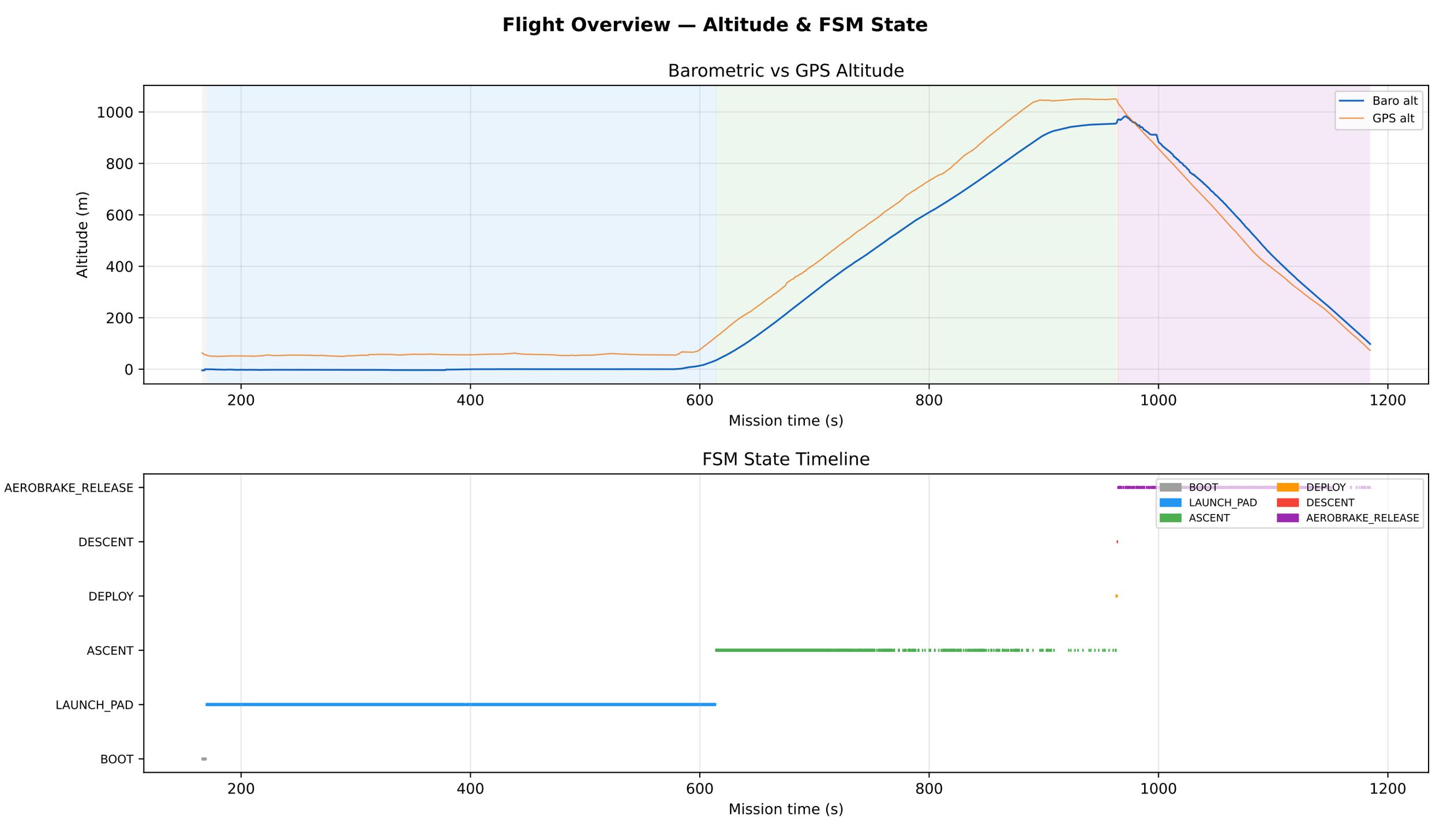
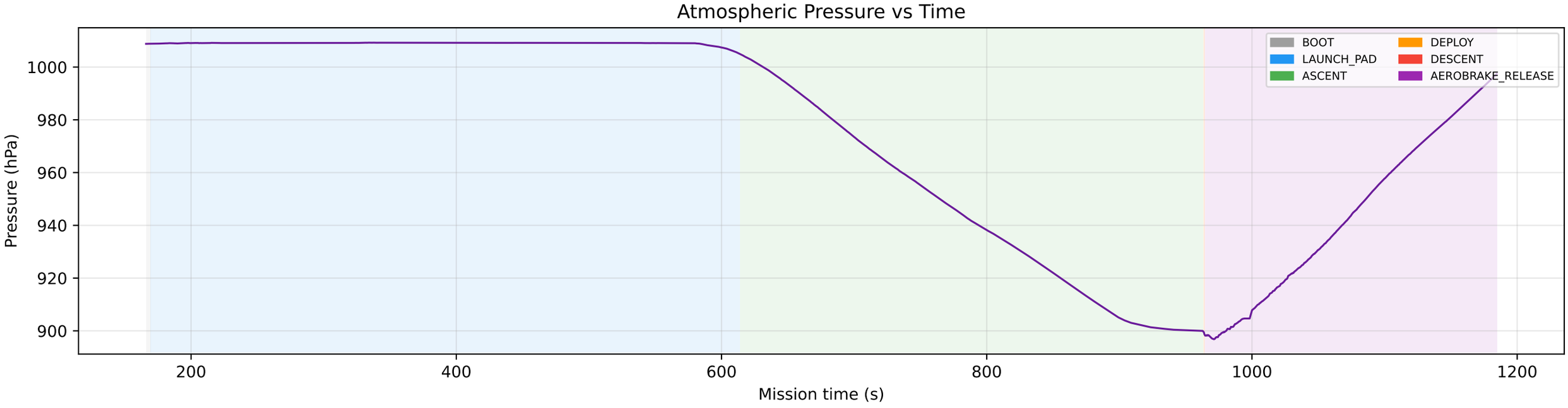
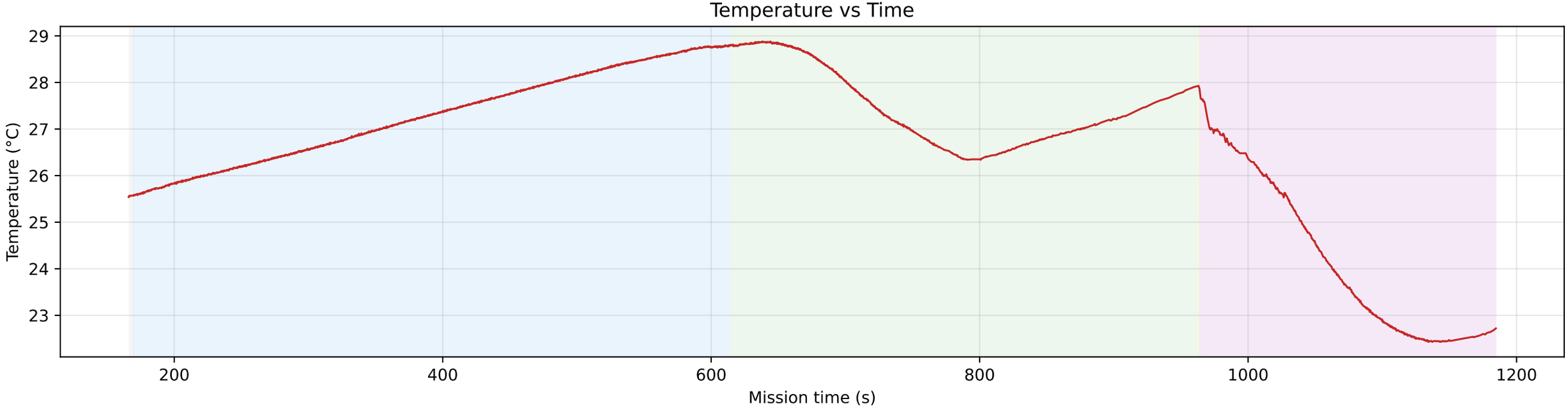
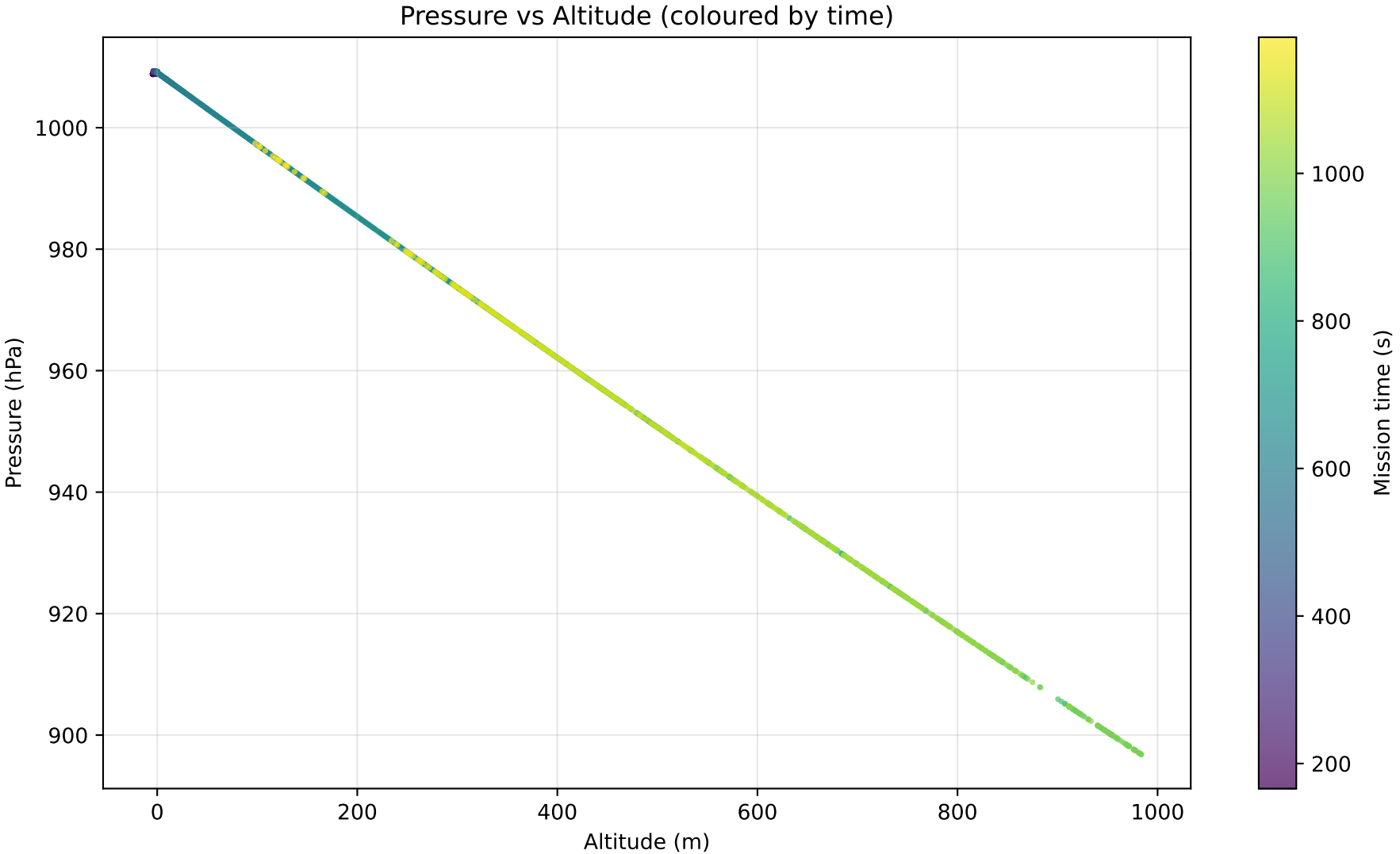
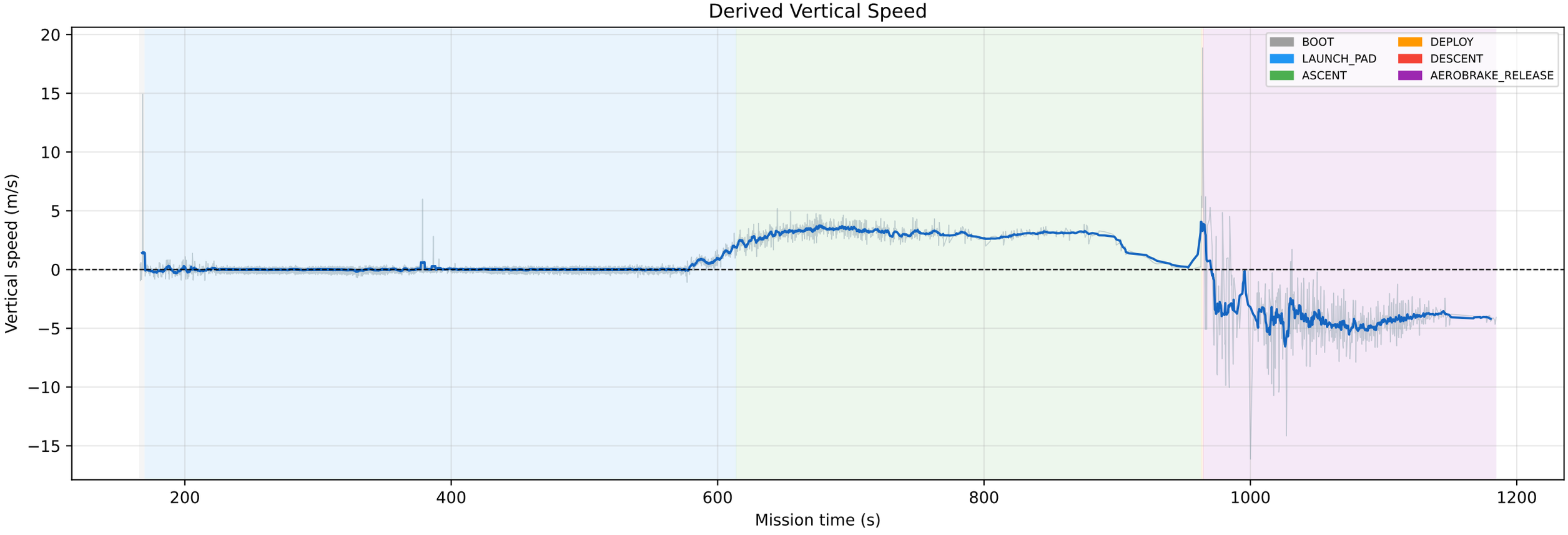

Flight Overview — Altitude & FSM State
Barometric vs GPS Altitude
Baro alt
GPS alt
Altitude (m)
0
200
400
600
800
1000
200
400
600
800
1000
1200
Mission time (s)
FSM State Timeline
AEROBRAKE_RELEASE
DESCENT
DEPLOY
ASCENT
LAUNCH_PAD
BOOT
BOOT
LAUNCH_PAD
ASCENT
DEPLOY
DESCENT
AEROBRAKE_RELEASE
200
400
600
800
1000
1200
Mission time (s)

Atmospheric Sensors

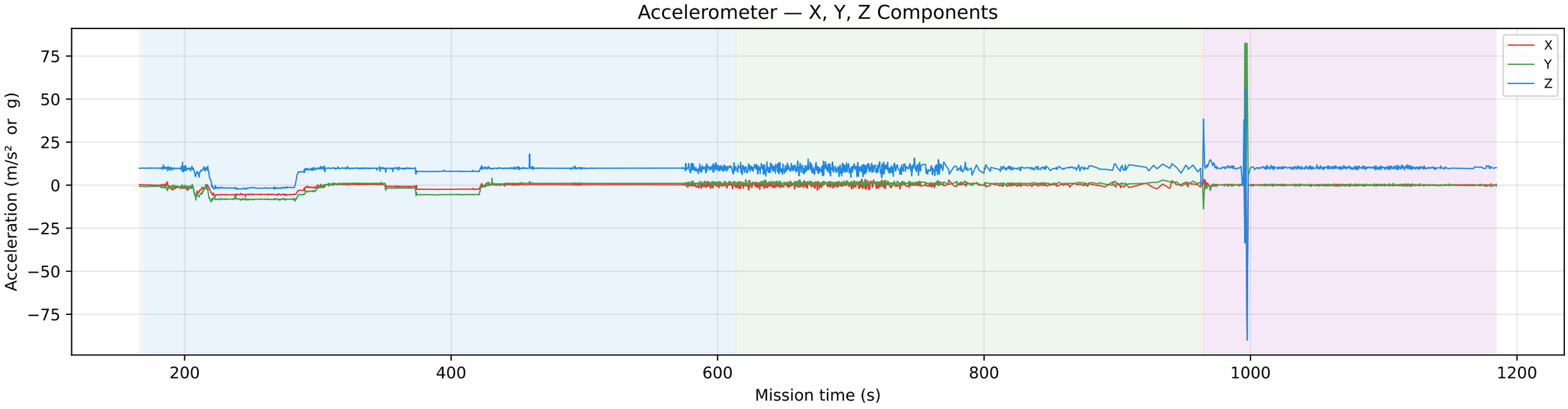


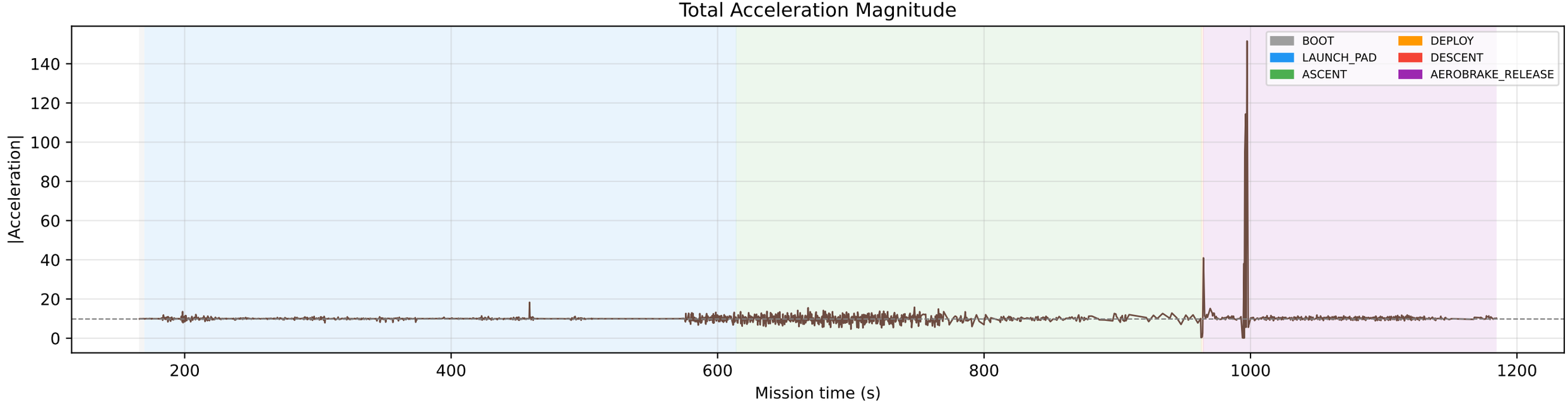



Pressure vs Altitude (coloured by time)
Pressure (hPa)
1000
980
960
940
920
900
0
200
400
600
800
1000
Altitude (m)
Mission time (s)
1000
800
600
400
200

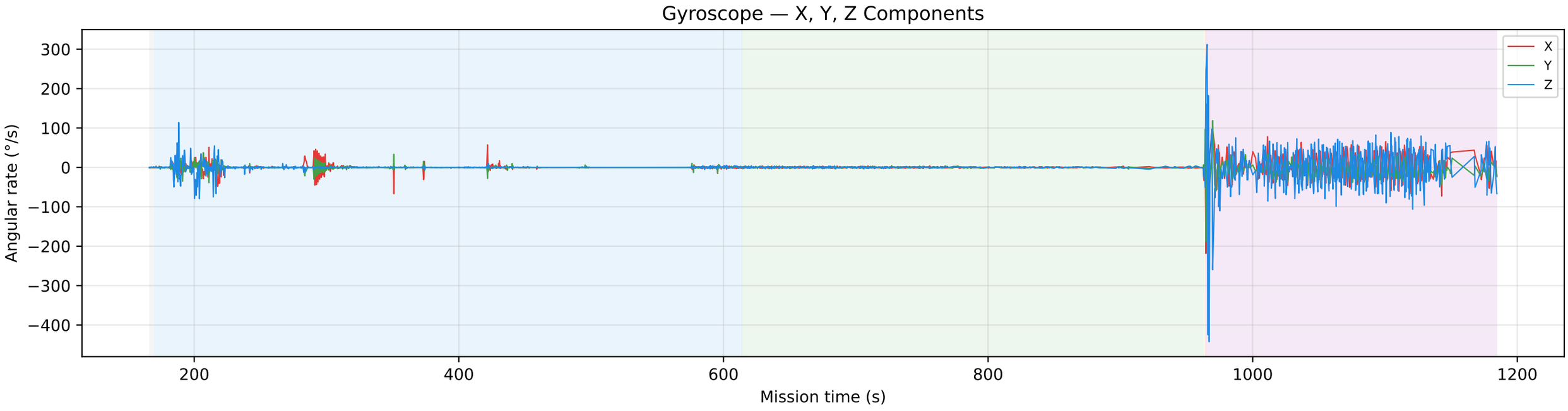

Derived Vertical Speed
BOOT
LAUNCH_PAD
ASCENT
DEPLOY
DESCENT
AEROBRAKE_RELEASE
Vertical speed (m/s)
20
15
10
5
0
−5
−10
−15
200
400
600
800
1000
1200
Mission time (s)

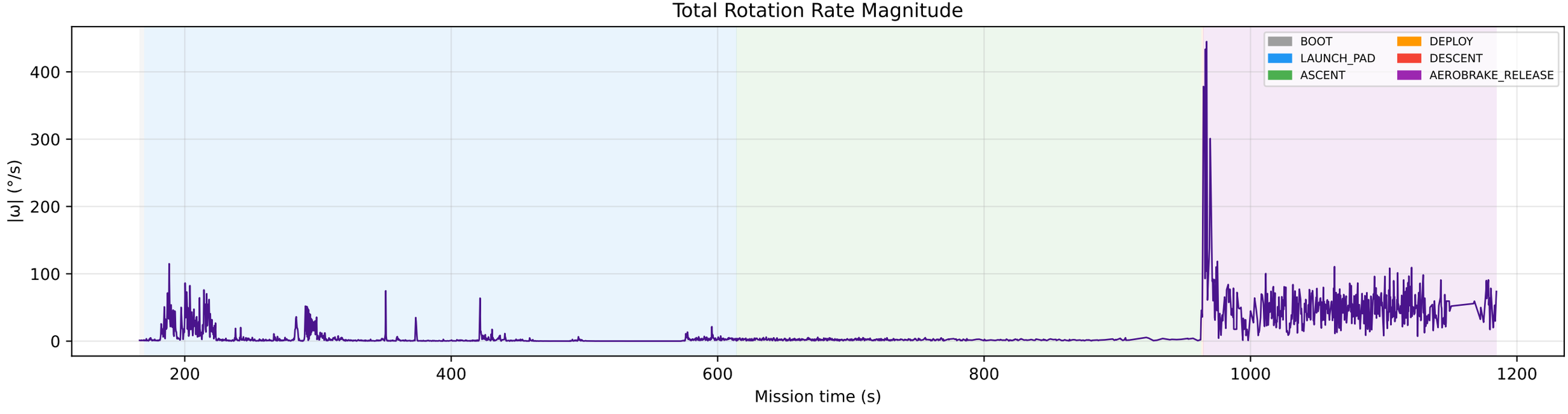
Accelerometer



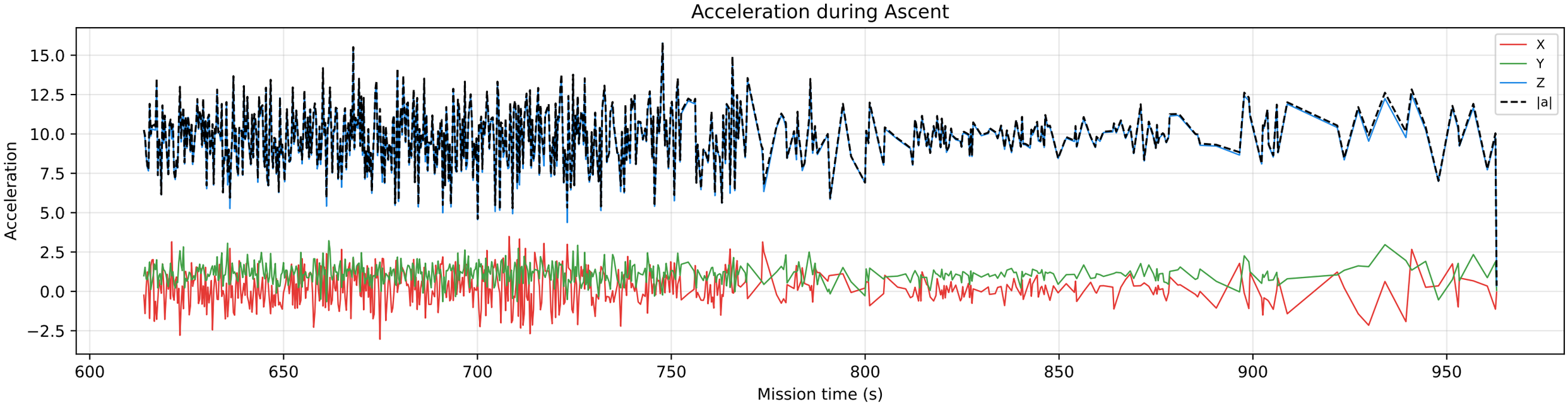


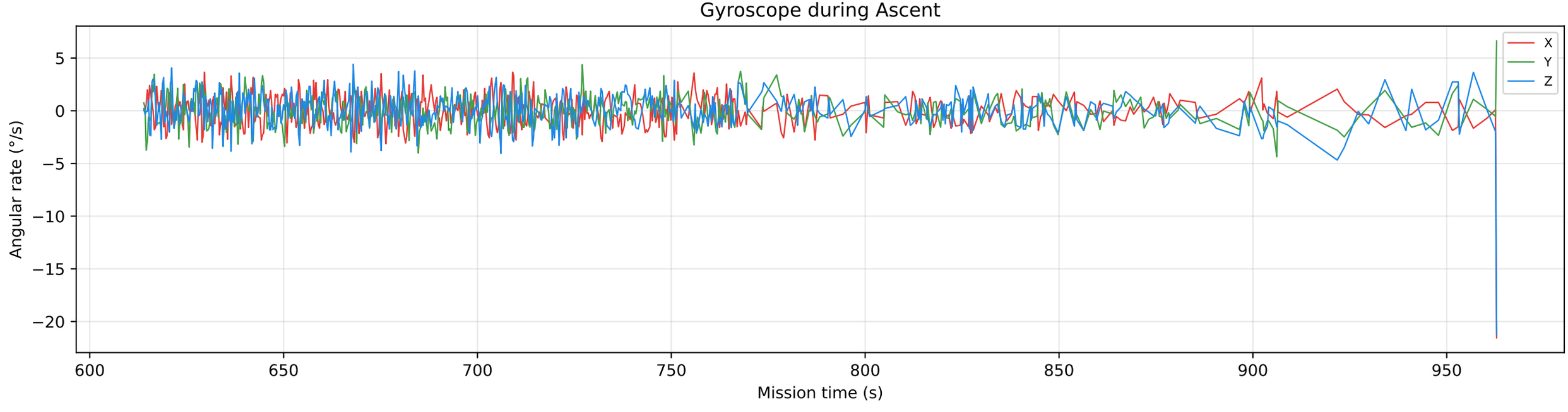
Gyroscope

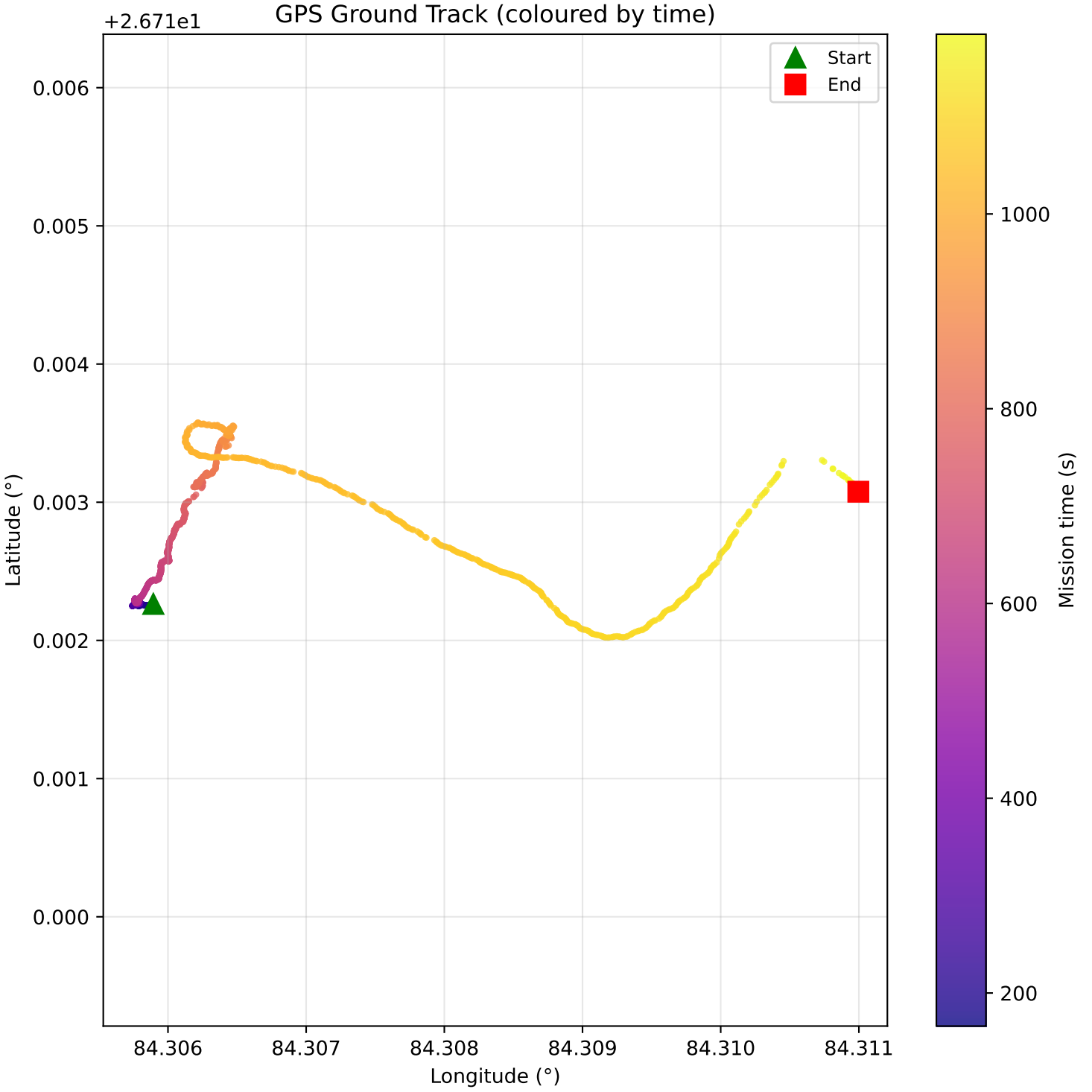

Gyroscope — X, Y, Z Components
Angular rate (°/s)
300
200
100
0
−100
−200
−300
−400
X
Y
Z
200
400
600
800
1000
1200
Mission time (s)

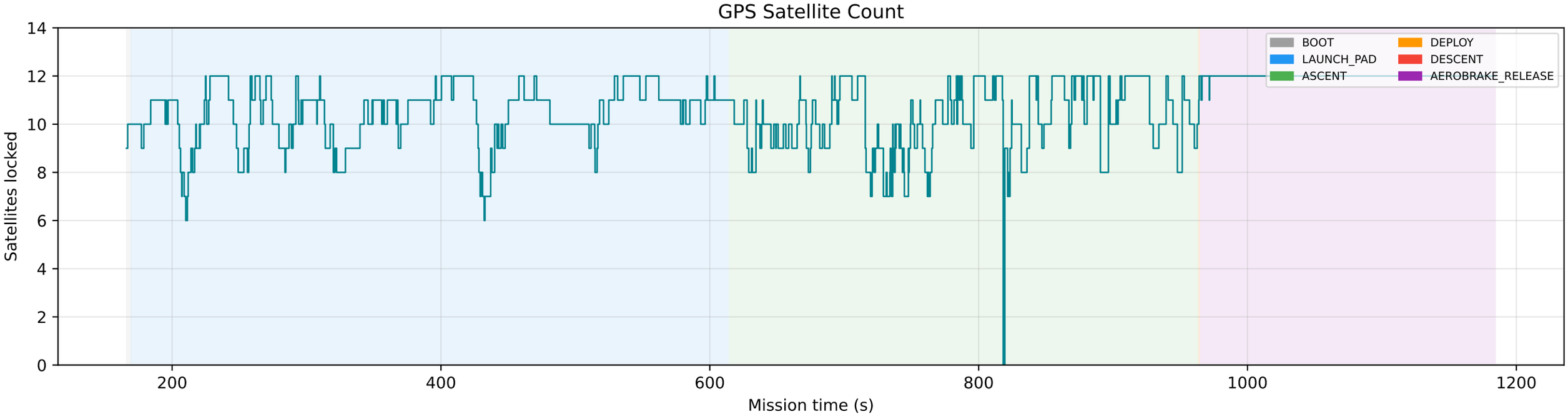

Total Rotation Rate Magnitude
|ω| (°/s)
400
300
200
100
0
BOOT
LAUNCH_PAD
ASCENT
DEPLOY
DESCENT
AEROBRAKE_RELEASE
200
400
600
800
1000
1200
Mission time (s)

IMU — Ascent Phase Zoom

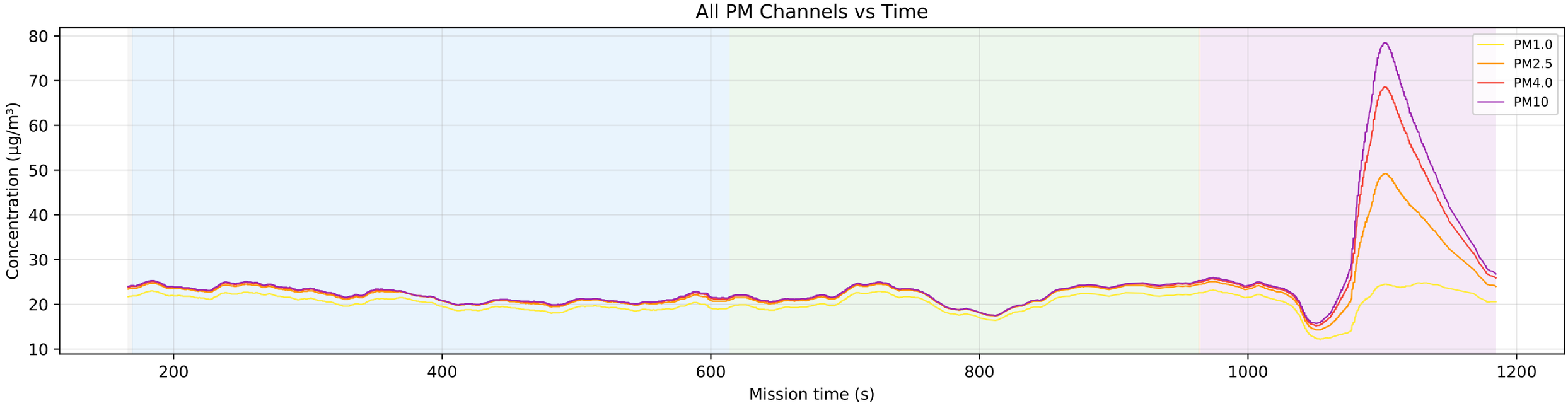


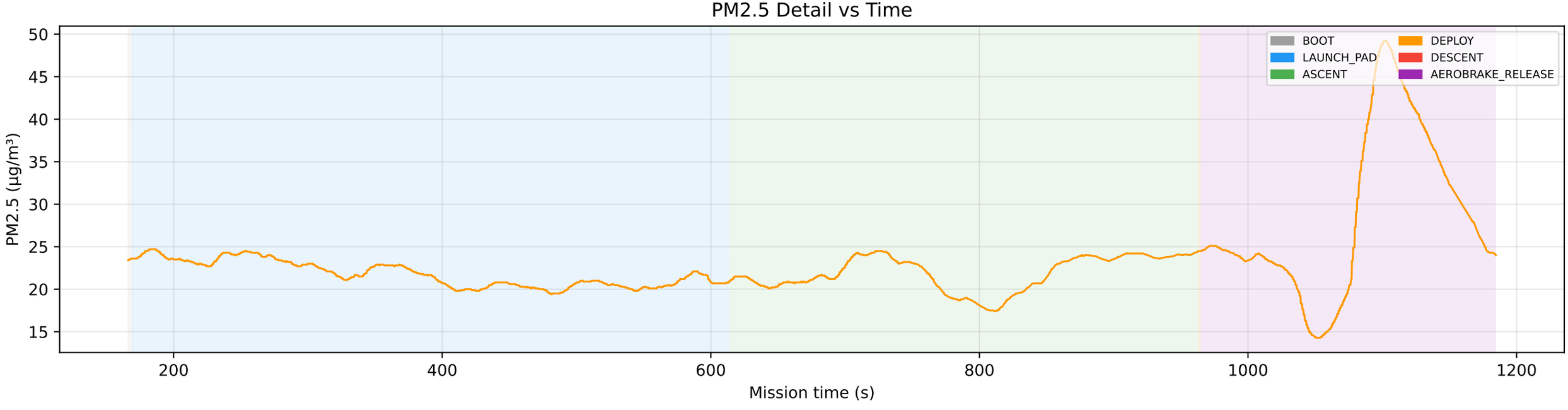


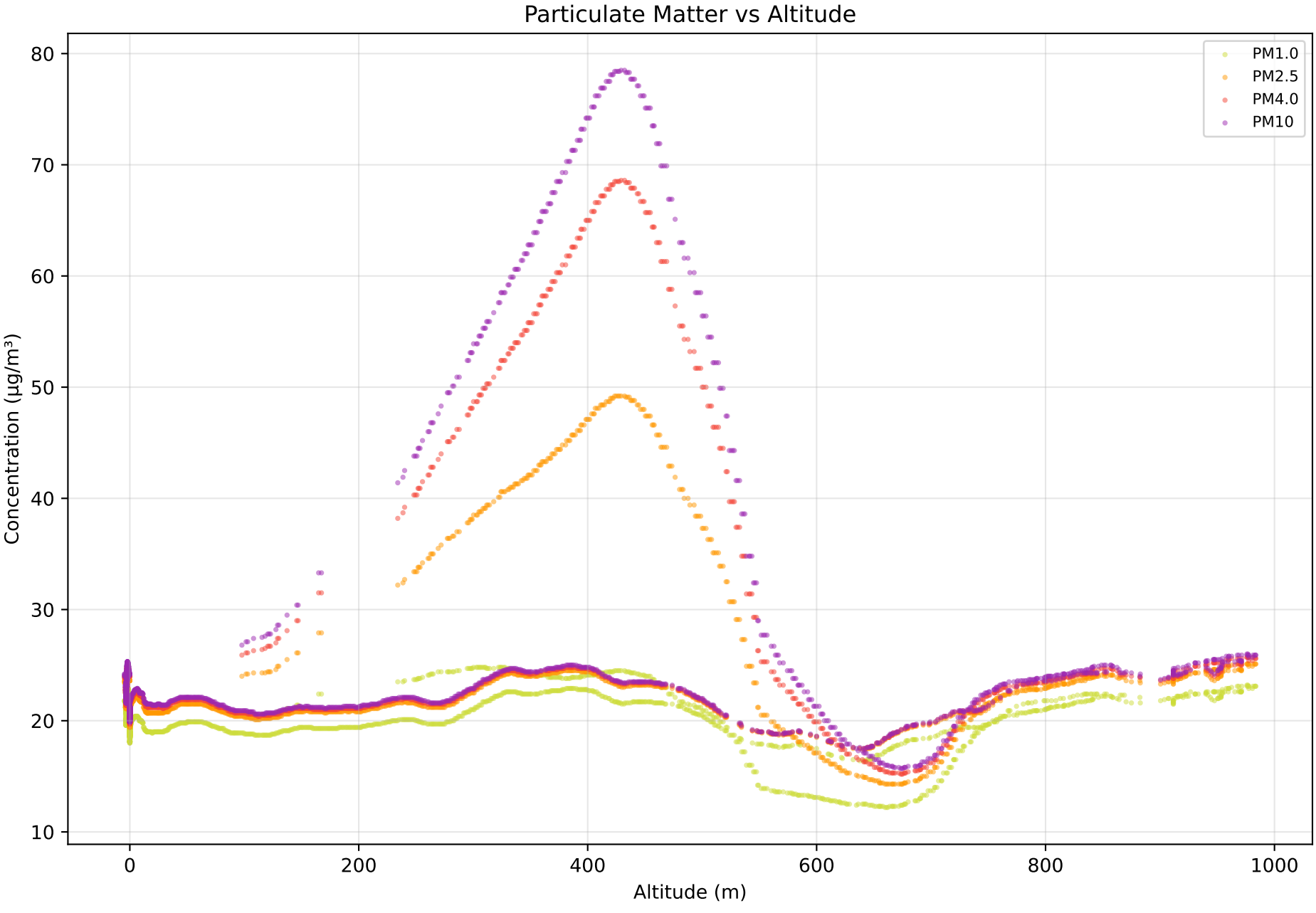

GPS Ground Track (coloured by time)
+2.671e1
Start
End
0.006
0.005
0.004
0.003
0.002
0.001
0.000
Latitude (°)
84.306
84.307
84.308
84.309
84.310
84.311
Longitude (°)
1000
800
600
400
200
Mission time (s)

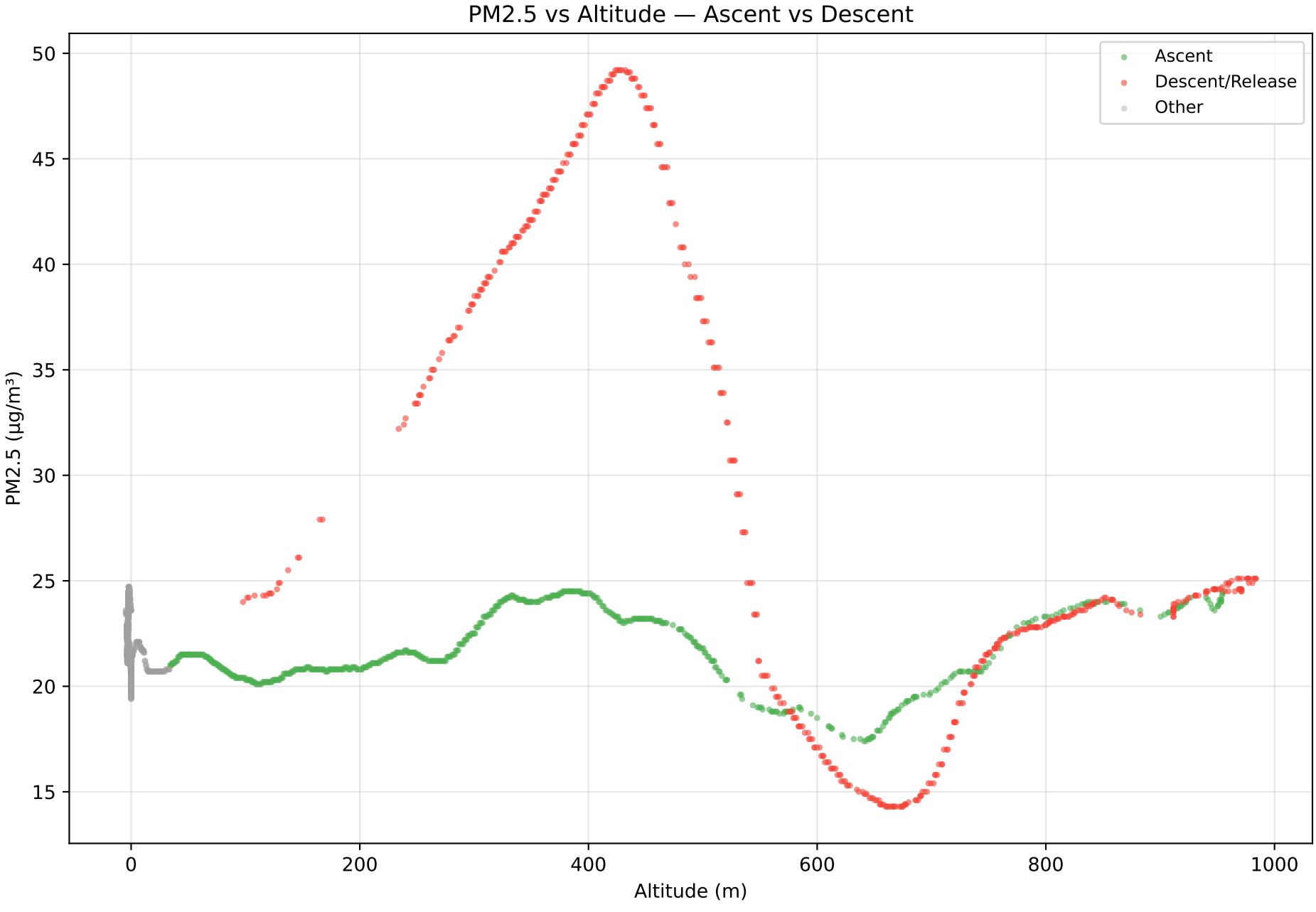

GPS Satellite Count
Satellites locked
Mission time (s)
0
2
4
6
8
10
12
14
200
400
600
800
1000
1200
BOOT
LAUNCH_PAD
ASCENT
DEPLOY
DESCENT
AEROBRAKE_RELEASE

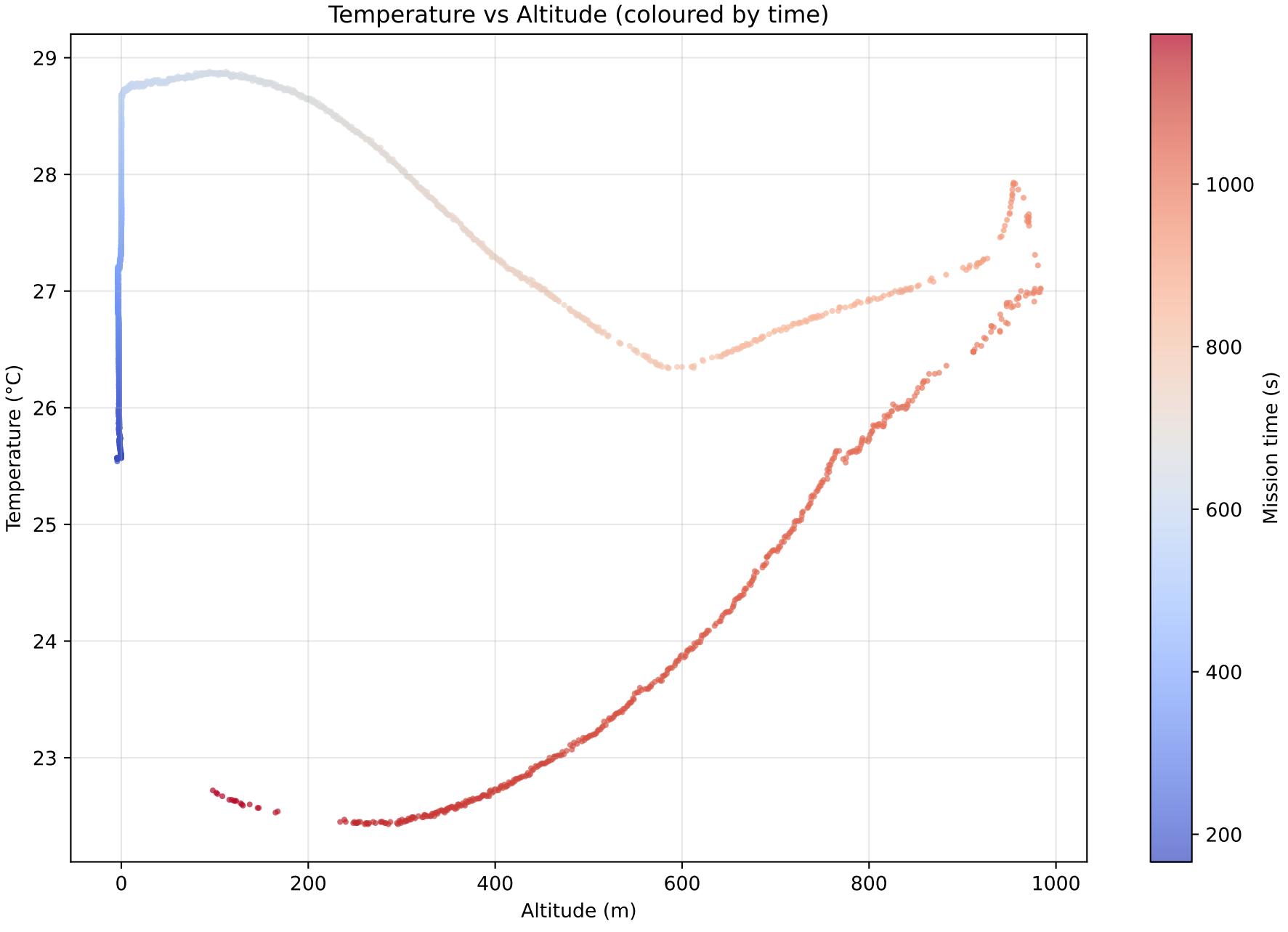
Particulate Matter vs Time

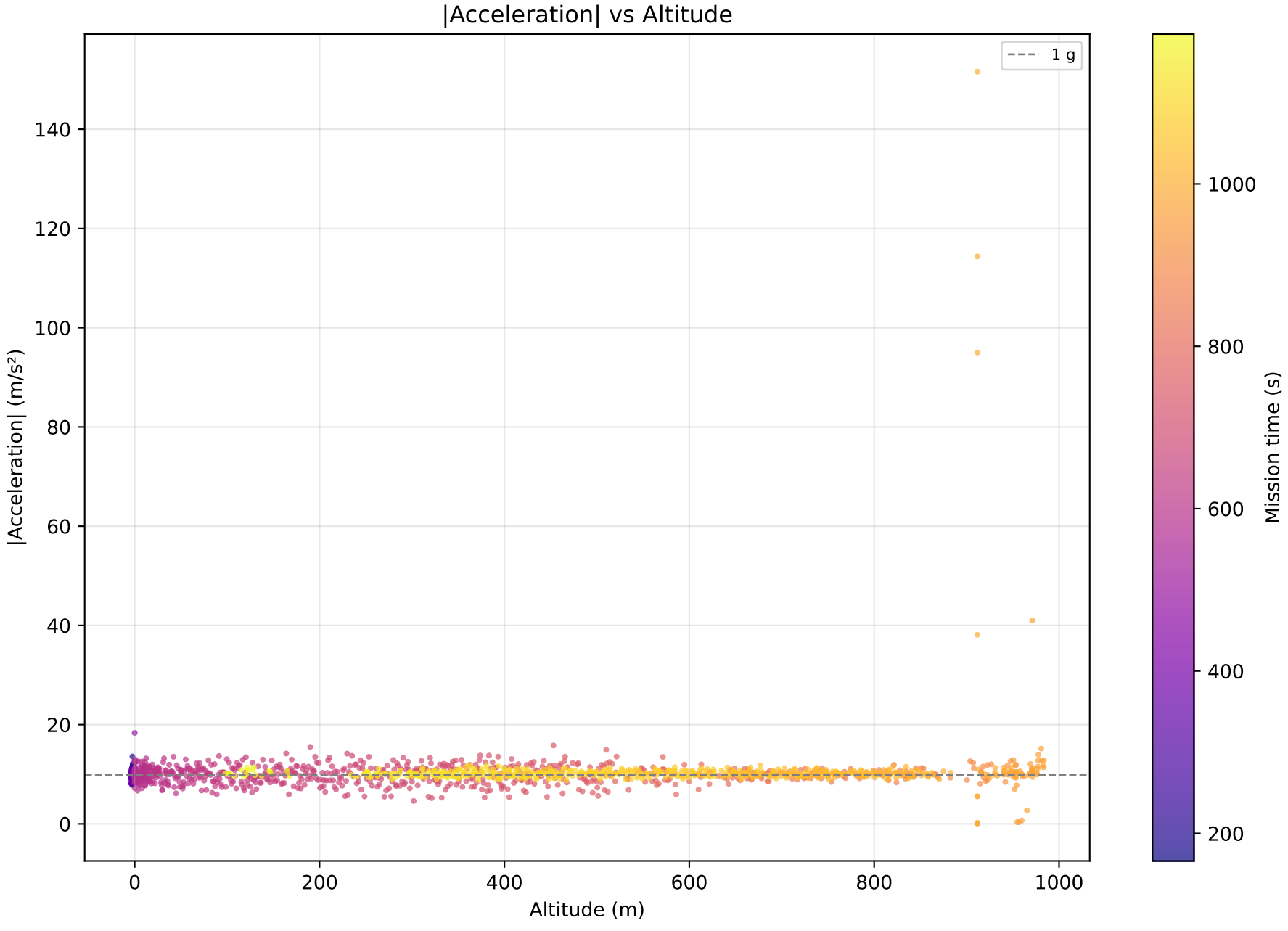


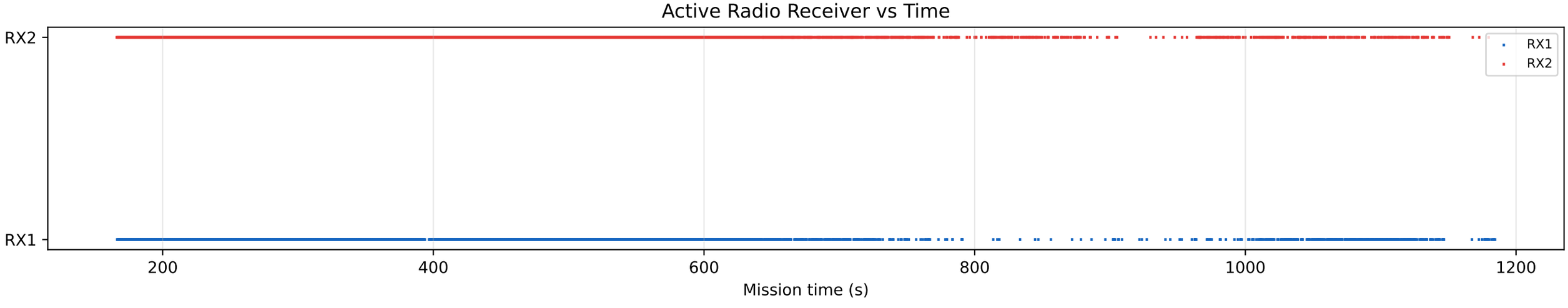



Particulate Matter vs Altitude
PM1.0
PM2.5
PM4.0
PM10
Concentration (µg/m³)
Altitude (m)
80
70
60
50
40
30
20
10
0
200
400
600
800
1000


PM2.5 vs Altitude — Ascent vs Descent
Ascent
Descent/Release
Other
PM2.5 (µg/m³)
Altitude (m)
50
45
40
35
30
25
20
15
0
200
400
600
800
1000


Temperature vs Altitude (coloured by time)
Temperature (°C)
29
28
27
26
25
24
23
0
200
400
600
800
1000
Altitude (m)
Mission time (s)
1000
800
600
400
200


|Acceleration| vs Altitude
1 g
|Acceleration| (m/s²)
Altitude (m)
Mission time (s)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
200
400
600
800
1000


Active Radio Receiver vs Time
RX2
RX1
RX1
RX2
200
400
600
800
1000
1200
Mission time (s)

Flight Dashboard — 2025ASI001

